

Fatima Zahrae Ahannuk

Big Data & AI Engineering Student · Data Engineering & Machine Learning

Seeking a 4–6 month end-of-studies internship (PFE) starting February 2027

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📍 Tetouan, Morocco



PROFILE

Final-year Big Data & AI engineering student comfortable across the whole data lifecycle: ETL pipelines with **Apache Spark**, SQL / NoSQL databases, **Machine Learning** models (classification, recommendation, NLP), **RAG** architectures and API deployment (**FastAPI**, Flask, **Docker**). I want to turn raw data into AI products used in production.

YEAR-END ENGINEERING PROJECT (PFA)

AUTO+ Morocco — Car Repair Shop Booking Platform

Year-end project (PFA), team of two · ENSA Tetouan · faculty-supervised

- Real data: scraping + OpenStreetMap → **123 garages** in **PostgreSQL/PostGIS**; **Redis** → **Parquet** pipeline; Node.js API (JWT, 3 roles).
- **Used-car price estimation**: **XGBoost** + Optuna on 70,000 listings → **MAE 11,800 MAD, R^2 0.94**.
- **Garage recommendation** (e5 embeddings, 87.5% top-1 + SVD) and fake-review detection, served with **FastAPI**.
- **AI diagnosis agent (RAG)**: OBD fault knowledge base in **pgvector**, Mistral LLM, guardrails; React Native app.

Python · XGBoost · RAG · pgvector · PostGIS · Redis · FastAPI · Node.js · React Native

EXPERIENCE

Explorer Program — UM6P (Mohammed VI Polytechnic University)

Selective innovation & entrepreneurship program

- Co-designed an AI-powered EdTech startup with a team: ideation, business model, mentoring and pitching to a jury.

PROJECTS

SmartLearn — Learning Platform with Recommendations

github.com/fahan860/smartlearn-platform

- **Spark / Pandas** ETL pipeline over 15,000 interactions (1,200 users, 150 courses); storage in **MongoDB** + **MySQL**.
- **FastAPI** recommendation microservice using **sentence-transformers embeddings**, with popularity-based fallback.
- 5 services containerized with **Docker Compose** (React, Express API, ML service, 2 databases).

Spark · Pandas · sentence-transformers · FastAPI · MongoDB · MySQL · Docker

Fake vs Real Face Detection — Deep Learning

github.com/merouane01/DL1

- 4 **CNN** architectures compared in **PyTorch** (VGG16, AlexNet, DenseNet-121 with transfer learning, custom CNN): **~87% accuracy**.

PyTorch · CNN · Transfer Learning · Computer Vision

Fake News Detection — NLP

github.com/fahan860/fake_news_detection

- TF-IDF + 4 models compared: **SVM reached 85.8% accuracy**; app served through a **Flask API**.

Scikit-learn · NLP · TF-IDF · Flask

Banking Anti-Overload System — Oracle PL/SQL

github.com/fahan860/bank-app-flask-oracle

- Operation queue processed by **PL/SQL triggers and procedures** with conflict detection; Flask monitoring.

Oracle · PL/SQL · Python · Flask

EDUCATION

National School of Applied Sciences of Tetouan (ENSA)

Engineering Degree — Big Data & Artificial Intelligence
2022 – 2027 (expected)

SKILLS

Languages

Python, SQL

Data Engineering

Apache Spark, ETL, Pandas, NumPy, Parquet, Redis

Databases

Oracle (PL/SQL), MySQL, PostgreSQL / PostGIS, MongoDB

ML & Deep Learning

Scikit-learn, XGBoost, SVM, Random Forest, recommender systems, PyTorch, CNN, transfer learning

NLP & Generative AI

Text vectorization, embeddings, LLMs, RAG, pgvector

Deployment

FastAPI, Flask, REST APIs, JWT, Docker, Git

BI

Power BI, Excel

CERTIFICATIONS

Intermediate Deep Learning with PyTorch — DataCamp

BCG X — GenAI Job Simulation (AI & Data Analysis) · 2026

Harnessing the Power of Data with Power BI — Microsoft · 2026

Master PL/SQL — EDUCBA (Coursera) · 2026

Introduction to Cloud — IBM · 2026

LANGUAGES

Arabic	Native
French	Fluent
English	Intermediate
Spanish	Beginner